Man Yeung (Andy) Tai, PhD

Postdoctoral Researcher • Federated Learning and Medical AI

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Summary

Machine learning researcher building privacy-preserving, federated clinical AI at the Institute for AI in Medicine (IKIM), University Hospital Essen. PhD in Neuroscience (UBC) applying machine learning to overdose risk prediction. 13 peer-reviewed journal articles (6 first-author; 412 citations), an accepted NeurIPS 2026 workshop paper, a CRAN package, and two years as a primary instructor in UBC's Master of Data Science.

Experience

Postdoctoral Researcher | IKIM, University Hospital Essen, Germany — Jul 2026 – Present

- Build federated chronic kidney disease risk prediction across general practices with Flower, patient-level DP-SGD, and SecAgg+ (FLIP-IT, EU/NRW-funded); jointly supervised by Prof. Jens Kleesiek and Prof. Michael Kamp.
- Develop privacy-preserving federated continual learning methods and an open-source benchmark toolkit, so clinical models learn from distributed hospital data without catastrophic forgetting.
- Investigate federated protocols that transmit activation and rank codes instead of parameters, with pre-registered controls and explicit communication-bitrate accounting.

Postdoctoral Teaching and Learning Fellow | UBC Master of Data Science and Statistics — Jul 2024 – Jun 2026

- Primary instructor for 10 courses reaching 550+ students, including Data Visualization (DSCI 531), Probability (DSCI 551, STAT 302), Databases and SQL (DSCI 513), and Communication and Argumentation (DSCI 542).
- Supervised 5 industry and clinical capstone teams (BC Children's Hospital, Vaccine Evaluation Center, Creative Destruction Lab, UBC Cybersecurity, Nautical Crime Investigation Services); the vessel-trajectory project was published in *Engineering Applications of AI* (2026; corresponding author).
- Co-authored *The Regression Cookbook* (UBC OER Fund); built the statistical framework (cumulative link mixed models) for a TLEF-funded AI literacy study; designed PrairieLearn assessments and Quarto course sites.

Data Science Consultant (freelance, part-time) | Clause Technology — Jul 2025 – Oct 2025

- Built HAVA, a maritime criminal detection system with NCIS: automated web scraping, NLP incident classification, and LLM-assisted de-duplication (BERT, NER, fuzzy matching) on a Dockerized MongoDB/PostgreSQL stack.
- Prototyped a computer vision and OCR pipeline for structured extraction from complex PDF tables.

Data Scientist | Concussion RX, Vancouver — Apr 2024 – Sep 2024

- Applied machine learning and statistical modelling to concussion subtype analysis; wrote clinical onboarding materials.

Graduate Researcher (PhD) | Addiction and Concurrent Disorders Group, UBC — Sep 2019 – Apr 2024

- Built overdose risk models (RF, XGBoost, SVM; N = 36,679; 91.1% AUROC) for RAMP, a CAD 1.4M Health Canada platform (co-investigator).
- First-authored a systematic review and meta-analysis of clinical decision support in addiction care (2024); keynotes at the 2nd International Psychiatry Congress (Egypt) and Virtual Solutions for Substance Use Care; visiting researcher at the University of Sydney Brain and Mind Centre (Mitacs Globalink).

Education

PhD, Neuroscience | University of British Columbia (fast-track transfer from MSc) — 2019 – 2024

Dissertation: *Machine Learning Applications in Addiction Risk Prediction*. Supervisor: Dr. Reinhard Michael Krausz.

Honours BSc, Neuroscience | University of Toronto; minors in Environmental Science and Religion — 2012 – 2017

Selected Publications and Software

- **Tai, A.M.Y.**, Kallenberg, N., Kleesiek, J., Kamp, M. (2026). Measuring Severity. *AXIOM 2026: Foundations of Efficient Deep Learning* (NeurIPS 2026 workshop), Grand Challenges track. Accepted.
- Sanghera, A., …, **Tai, A.M.Y.** (corresponding) (2026). Semi-supervised vessel trajectory analysis for unregulated fishing activity detection. *Engineering Applications of Artificial Intelligence*, 169, 113933.
- **Tai, A.M.Y.**, et al. (2026). Rethinking AI in youth mental health: promise, perils, and ethical integration. *International Journal of Mental Health Systems*, 20(1).
- **Tai, A.M.Y.**, et al. (2019). Machine learning and big data: Implications for disease modeling and therapeutic discovery in psychiatry. *Artificial Intelligence in Medicine*, 99, 101704.
- **GGenemy** (CRAN R package): audits ggplot2 figures for accessibility (WCAG 2.1) and misleading design.

Skills

ML and statistics: PyTorch, TensorFlow, scikit-learn, BERT/transformers, NLP; mixed-effects models, meta-analysis
Federated learning: Flower; privacy-preserving, communication-efficient distributed and continual learning
Programming and data: Python, R, SQL, JavaScript; PostgreSQL, MongoDB; Docker, Git, AWS (basics); LiteLLM
Languages: English (native), Cantonese (fluent), Mandarin (conversational), German (beginner)

Awards and Service

Awards: Mitacs Globalink Research Award (CAD 6,000, 2022–23); President's Academic Excellence Initiative PhD Award (2021–23); UBC MDS Teaching Assistant Award (2021–22); Faculty of Medicine Graduate Award (2019).
Service: 37 peer reviews for 11 journals (incl. *JMIR*, *JMIR AI*); UBC Statistics EDI seminar coordinator.